

1. Explain the PEAS description for the vacuum cleaner world problem.

In AI, when we design an intelligent agent, we define it using the **PEAS** framework. This stands for **P**erformance, **E**nvironment, **A**ctuators, and **S**ensors.

Here is the PEAS description for a simple automated vacuum cleaner:

- **Performance Measure (P):** How do we know if the robot is doing a good job?
 - **Cleanliness:** It gets points for cleaning up dirt.
 - **Efficiency:** It might lose points for every move it makes (to save battery) or for taking too long.
 - **Safety:** It must not fall down stairs or damage the floor.
- **Environment (E):** Where does the robot operate?
 - A grid of squares (often simplified to just two squares, A and B).
 - The squares can be either **Dirty** or **Clean**.
- **Actuators (A):** How does the robot act or move?
 - **Motors/Wheels:** To move *Left* or *Right*.
 - **Suction:** To *Suck* up the dirt.
 - **Idle:** To wait and do nothing.
- **Sensors (S):** How does the robot see the world?
 - **Location Sensor:** Tells the robot which square it is in (e.g., "I am in square A").
 - **Dirt Sensor:** Detects if the current square is dirty.

2. Give the difference between informed search and uninformed search algorithms.

Think of searching for a location on a map. These are the two main ways an AI can search for a solution.

Basis	Uninformed Search	Informed Search
1. Meaning	Does not use any extra information about goal	Uses extra information (heuristic) to find goal
2. Also called	Blind Search	Heuristic Search
3. Knowledge used	Only initial state and rules	Uses knowledge about how close we are to goal
4. Speed	Usually slow	Usually faster
5. Efficiency	Less efficient	More efficient
6. Path selection	Tries all possible paths	Chooses best path first

Basis	Uninformed Search	Informed Search
7. Goal direction	No direction (searches blindly)	Searches toward goal
8. Example algorithms	BFS, DFS, Uniform Cost Search	A*, Greedy Best First Search
9. Memory usage	Can use more memory	Usually optimized
10. Real life example	Searching a key randomly in room	Searching key where you think you kept it

3. State any four applications of AI.

AI is used in almost every industry today. Here are four common applications:

1. Healthcare (Medical Diagnosis):

- AI programs analyze medical images (like X-rays) to detect diseases such as cancer earlier and more accurately than humans. They can also suggest personalized treatment plans.

2. Finance (Fraud Detection):

- Banks use AI to monitor credit card transactions in real-time. If it sees a suspicious payment (e.g., a large purchase in a foreign country), it flags it as fraud instantly.

3. Transport (Self-Driving Cars):

- Vehicles from companies like Tesla or Waymo use AI to perceive the road, obey traffic signals, avoid pedestrians, and drive without human intervention.

4. Natural Language Processing (Chatbots):

- Virtual assistants like Siri, Alexa, and customer service chatbots use AI to understand human speech and text, allowing them to answer questions and obey commands.

4. Demonstrate the 4 ways we can measure performance of AI Algorithms.

When we judge how good a search algorithm is, we look at these four specific metrics:

1. Completeness:

- *Does it guarantee a solution?*
- If a solution exists, a complete algorithm will definitely find it. It won't get stuck in a loop forever or give up.

2. **Optimality:**

- *Does it find the best solution?*
- If there are multiple ways to solve a problem, an optimal algorithm finds the one with the lowest cost (shortest path, least money, or fastest time).

3. **Time Complexity:**

- *How long does it take?*
- This measures the speed of the algorithm. We want an algorithm that finds the answer quickly rather than taking years to calculate.

4. **Space Complexity:**

- *How much memory does it need?*
- This measures how much computer memory (RAM) the algorithm needs to store information while it is searching.

5. **List components of AI program.**

An AI program generally consists of three major components that allow it to function intelligently:

[Knoladegbase](#) , [Resining](#),
[learning](#), [ui](#)

1. **Knowledge Base (The "Library"):**

- This contains the facts, rules, and data about the domain. It is what the AI "knows."
- *Example:* "If a patient has a fever, they might have the flu."

2. **Inference Engine (The "Brain"):**

- This is the reasoning part. It applies logic to the data in the Knowledge Base to deduce new information or make decisions.
- *Example:* It sees a patient has a fever (input) and uses the rule above to suggest "Flu" (output).

3. **User Interface (The "Mouth/Ears"):**

- This allows humans to interact with the system. It takes the user's query and presents the AI's answer in a readable format.

*(Note: In the context of Intelligent Agents specifically, the components are often listed as **Sensors** (input), the **Agent Function** (processing/logic), and **Actuators** (output).)*

6. Describe Stochastic and Episodic task environments.

When analyzing the environment an AI works in, we classify it based on how complex or predictable it is.

- **Stochastic Environment (Random):**
 - **Definition:** In this environment, the outcome of an action is **not** completely predictable. Even if the AI does the same thing twice, the result might be different because of random factors.
 - **Key Feature:** There is uncertainty.
 - **Example: Driving a car.** Even if you turn the wheel perfectly, traffic, weather, or a flat tire can change what happens next.
- **Episodic Environment (Isolated):**
 - **Definition:** The AI's life is chopped up into small, separate "episodes." What the AI decides to do in one episode **does not** affect the next one. The AI doesn't need to "plan ahead."
 - **Key Feature:** Past actions don't matter for the future.
 - **Example: A robot spotting defective parts on an assembly line.** Looking at part #1 has no effect on whether part #2 is defective. Each part is a fresh start.

Spam Email Filter

One email comes → decide spam or not spam

Next email comes → previous email doesn't matter



<https://www.geeksforgeeks.org/artificial-intelligence/components-of-artificial-intelligence-ai/>

7. List any two components of an AI program.

While Question 5 asked for all components, here are the two most critical ones simplified:

1. **Data (or Knowledge Base):** The information the program possesses. For example, a chess program needs to know the rules of the game and the positions of the pieces.
2. **Algorithm (or Reasoning Logic):** The code that tells the program *how* to use the data to solve the problem. For example, the logic that calculates the best next move in chess.

8. Define the concept of rationality in agents.

In AI, "Rationality" doesn't just mean "thinking like a human." It means **doing the right thing** to get the best result.

- **Definition:** A rational agent is one that chooses the action that is expected to maximize its **Performance Measure**, given the evidence provided by its **Percept Sequence** (everything it has perceived so far) and its built-in knowledge.
- **Simple Rule:** Rationality = Maximizing Success.
- **Important Note:** Rationality is **not** omniscience (knowing everything). An agent can be rational but still make a mistake if it didn't have enough information (e.g., crossing a street when a car was hidden around a blind corner).

9. Distinguish between Weak AI and Strong AI.

Basis	Weak AI	Strong AI
1. Meaning	AI designed for a specific task	AI with human-like intelligence
2. Also called	Narrow AI	General AI (AGI)
3. Intelligence level	Limited intelligence	Full human-level intelligence
4. Thinking ability	Cannot think or understand	Can think and understand like humans
5. Self-awareness	No self-awareness	Has self-awareness (theoretical)
6. Decision making	Works only as programmed	Can make its own decisions
7. Examples	Siri, Alexa, Google Maps	Not yet developed fully
8. Availability	Already exists	Still under research
9. Learning ability	Learns within limits	Can learn anything like humans
10. Consciousness	No consciousness	Would have consciousness

10. Describe an Intelligent Agent.

An Intelligent Agent is the central character in AI study. It is anything that interacts with its surroundings.

- **Definition:** An entity that **perceives** its environment through sensors and **acts** upon that environment through actuators to achieve a goal.
- **The Cycle:**
 1. **Perceive:** The agent uses sensors (cameras, microphones) to gather data from the world.
 2. **Think/Process:** The agent function (brain) decides what to do based on that data.
 3. **Act:** The agent uses actuators (motors, screens, speakers) to change the environment.
- **Equation:**

Agent = Architecture + Agent Program

- **Architecture:** The physical hardware (the robot body or computer).
- **Program:** The software logic running inside.

Example

A self-driving car perceives the road using cameras and sensors and acts using steering, brakes, and acceleration.

11. Explain space and time complexity in search algorithms.

In AI, we don't just care if an algorithm finds the answer; we care about how "expensive" it is to run.

- **Time Complexity:**
 - **Definition:** This measures how much **time** it takes to find a solution.
 - **How it's measured:** We count the number of nodes or states the algorithm has to explore before reaching the goal.
- **Space Complexity:**
 - **Definition:** This measures how much **memory (RAM)** the algorithm needs.
 - **How it's measured:** We look at the maximum number of nodes the algorithm has to keep in its memory at any single time during the search.

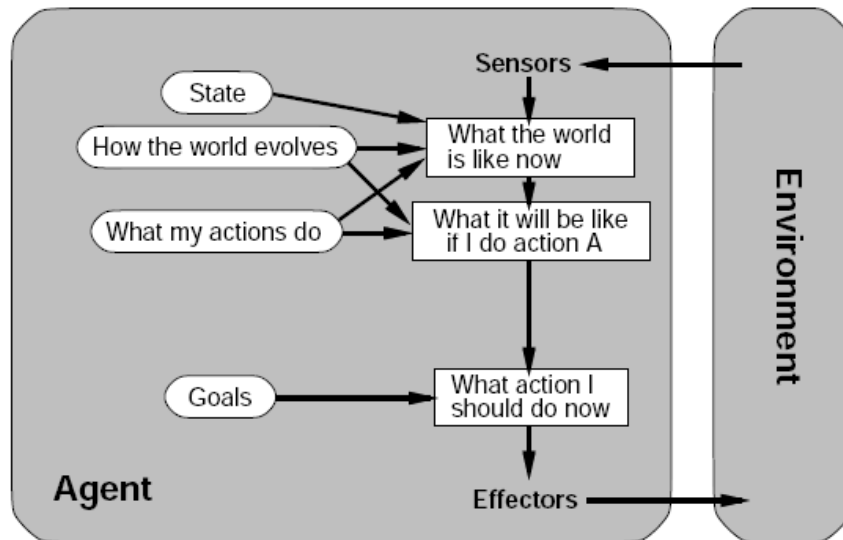
Why it matters: Some algorithms (like Breadth-First Search) find the shortest path but might crash your computer because they try to store too much data in the RAM (high space complexity).

12. Distinguish between a deterministic and a stochastic observable environment.

Basis	Deterministic Environment	Stochastic Environment
1. Meaning	Output is fixed and predictable	Output is random or uncertain
2. Result of action	Same action → Same result every time	Same action → Different results possible
3. Uncertainty	No uncertainty	Has uncertainty
4. Predictability	Fully predictable	Not fully predictable
5. Control	Easy to control	Hard to control
6. Mathematical nature	No probability involved	Uses probability
7. Example	Calculator, Chess game	Weather system, Stock market
8. Risk factor	No risk	Risk involved

14. Short note on: Goal-based agent with diagram.

A **Goal-based agent** is smarter than a simple "if-then" robot. Instead of just reacting to the moment, it has a **target** it wants to reach.



A **Goal-Based Agent** is an intelligent agent that works to achieve a specific goal.

It does not just react to the environment.
Instead, it **plans actions** to reach a desired goal.

Key Points:

- It has a **goal** to achieve.
- It checks different possible actions.
- It selects the action that helps reach the goal.
- It may use search and planning techniques.
- It is more flexible than a simple reflex agent.

Example:

- **Google Maps** finding the best route to reach a destination.
- A robot planning steps to pick up an object.

15. Differentiate between a problem-solving agent and a learning agent.

Basis	Problem-Solving Agent	Learning Agent
1. Meaning	Solves a problem using given knowledge	Improves performance by learning from experience
2. Knowledge	Uses predefined rules and information	Gains new knowledge over time
3. Learning ability	Does not learn from past experience	Learns from past actions and feedback
4. Adaptability	Less adaptable	Highly adaptable
5. Working style	Searches for solution to reach goal	Observes, learns, and improves
6. Improvement	Performance remains same	Performance improves over time
7. Components	Initial state, goal state, search method	Learning element, performance element, critic, problem generator
8. Example	Solving a maze using fixed algorithm	Spam filter improving after seeing emails

17. Describe Problem Formulation steps. Formulate 8-tile Problem.

Problem Formulation is the process of deciding what actions and states to consider. The steps are:

1. **Initial State:** Where the agent starts.
2. **Actions:** What the agent can do in any state.
3. **Transition Model:** A description of what each action does (the result).
4. **Goal Test:** A way to check if the current state is the solution.
5. **Path Cost:** A numeric cost (like time or distance) assigned to each path.

Formulating the 8-tile Problem:

- **Initial State:** Any random arrangement of 8 numbered tiles on a 3×3 grid.
- **Actions:** Move the "blank" space *Left*, *Right*, *Up*, or *Down*.
- **Transition Model:** If we move the blank "Up," it swaps places with the tile above it.
- **Goal Test:** Check if the tiles are in the correct order (1, 2, 3, 4, 5, 6, 7, 8).
- **Path Cost:** Each move costs 1.

18. Explain A Algorithm with an example.*

Definition:

A* is an **informed search algorithm** used to find the shortest path from a start node to a goal node.

It uses both actual cost and heuristic cost to select the best path.

Evaluation Function:

$$f(n)=g(n)+h(n) \quad f(n) = g(n) + h(n) \quad f(n)=g(n)+h(n)$$

Where:

- **g(n)** = Actual cost from start node to current node
- **h(n)** = Estimated cost from current node to goal (heuristic)
- **f(n)** = Total estimated cost of solution through node n

Steps of A* Algorithm:

1. Start with the initial node.
2. Calculate f(n) for all successor nodes.
3. Select the node with the smallest f(n).
4. Expand that node.
5. Repeat the process until the goal node is reached.

Example:

Consider the following graph:

Start node = S

Goal node = G

Node g(n) h(n) f(n)

A	1	4	5
B	2	2	4
G	4	0	4

From S, two paths are available: A and B.

Calculate f(n):

- For A: $f(A) = 1 + 4 = 5$
- For B: $f(B) = 2 + 2 = 4$

Since B has smaller $f(n)$, choose B.

From B to G:

$$f(G) = 4 + 0 = 4$$

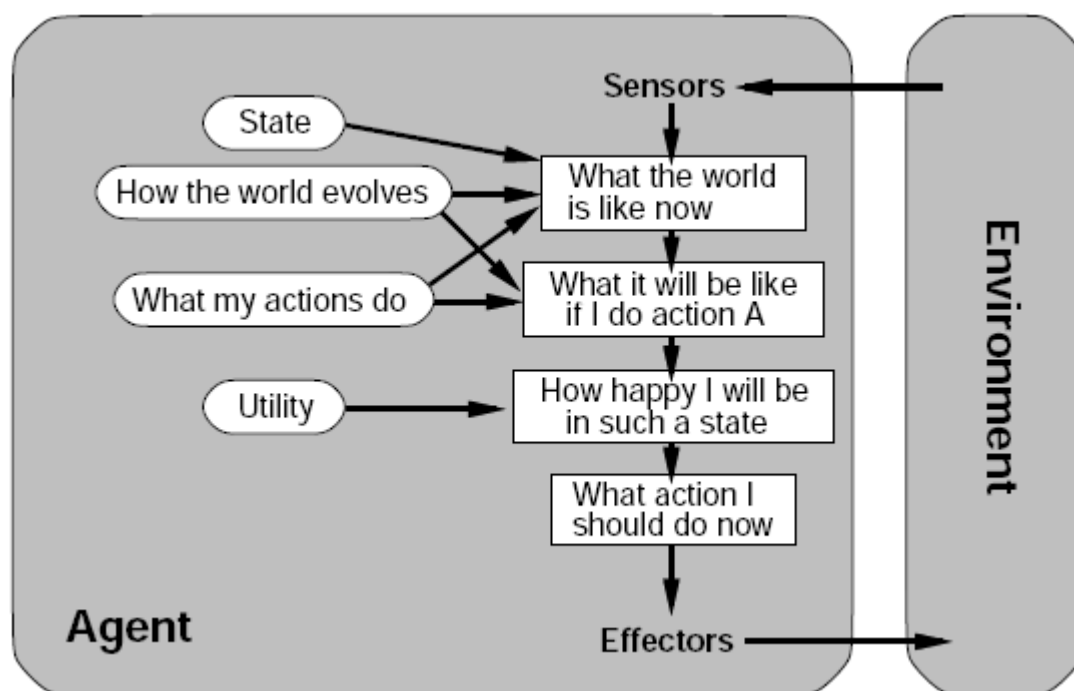
Thus, the shortest path is:

S → **B** → **G**

Advantages:

- Finds optimal solution if heuristic is admissible.
- More efficient than uninformed search.
- Widely used in pathfinding and games.

19. Explain utility-based model agent in detail.



Definition:

A **Utility-Based Agent** is an intelligent agent that chooses actions based on a **utility function**.

It selects the action that gives the **maximum utility (maximum benefit or happiness)**.

What is Utility?

Utility is a numerical value that shows how good or desirable a state is.
Higher utility = Better result.

Working of Utility-Based Agent:

1. Observes the current state.
2. Predicts possible outcomes of different actions.
3. Calculates utility value for each outcome.
4. Selects the action with **highest utility**.

Example:

- **Self-driving car** choosing the safest and fastest route.
- **Online shopping website** recommending products you may like.
- **Investment system** choosing the most profitable option.

20. Describe the categories of intelligent systems with examples.

Intelligent systems are generally categorized by what they "think" or "do."

1. Systems that Think Humanly

These systems try to imitate human thinking and reasoning process.

Example: Cognitive modeling systems.

2. Systems that Act Humanly

These systems behave like humans so that people cannot tell the difference.

Example: The Turing Test, chatbots.

3. Systems that Think Rationally

These systems use logical rules to reach correct conclusions.

Example: Expert systems (medical diagnosis system).

4. Systems that Act Rationally

These systems choose actions that give the best possible result.
Example: Self-driving cars, chess-playing AI.